THE OHIO STATE UNIVERSITY

College of Arts and Sciences

School of Earth Sciences

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May 2, 2025

To: Professor Joachim Moortgat, PhD
School of Earth Sciences, The Ohio State University
275 Mendenhall Laboratory, 125 South Oval Mall
Columbus, OH 43210-1398

Dear Joachim,

I enthusiastically support the creation of the BuckAI Observatory—an applied-AI Center of Excellence in the College of Arts and Sciences. The Observatory will lower technical barriers, catalyze interdisciplinary collaborations, and position OSU as a national leader in AI-driven Earth observation, accelerating the analysis of vast datasets, identifying patterns and anomalies that might be missed by human analysts, and enabling more accurate and timely predictions and assessments of environmental changes.

Research in my lab focuses on critical zone processes (such as weathering, hydrology, and geochemistry) that connect Earth's surface and subsurface, particularly the interactions between land, water, vegetation, and human activities. This research deals with large datasets (e.g., forest fire data acquired by VIIRS and MODIS; infrared and microwave data used for monitoring ocean heat stress and coral bleaching events; gravity field data from GRACE satellites used to monitor changes in water storage on land), which are increasingly provided by remote sensing techniques, enabling large-scale, long-term monitoring of these critical zone processes. As such, I can clearly see the benefits that the BuckAI Observatory will bring to my ability to acquire and process large remote-sensing products and datasets.

I expect that a seed investment by ASC in shared storage and computational resources, matched with initial technical support (a data engineer and student assistance with manually generating training data), as well as the other initiatives outlined in your white paper (such as championing interdisciplinary teams and annual workshops) will increase my success in pursuing extramural funding from NSF (particularly through its Water, Landscape, and Critical Zone Processes (WaLCZ) program), the Critical Zone Collaborative Network (CZNet), and the US Department of Energy Environmental System Science Program (ESS).

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I am also interested in learning more about the opportunities to broaden my funding sources beyond federal funding agencies through partnerships with industry and private foundations as well through the commercialization of intellectual property developed by my team.

Beyond research, I endorse the proposed goal of better educating our students for a marketplace in which AI will play an increasingly critical role. I expect that students in my department will be interested in classes such as ES5757 "AI in Geosciences", and plan to either share domain-specific case studies and/or contribute to new AI-related courses in my own unit.

Should ASC support the founding of the BuckAI Observatory, please reach out to me about how I can be involved and/or for any further information that would be helpful.

Yours sincerely,

Ozeas S. Costa, Jr., Ph.D.
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